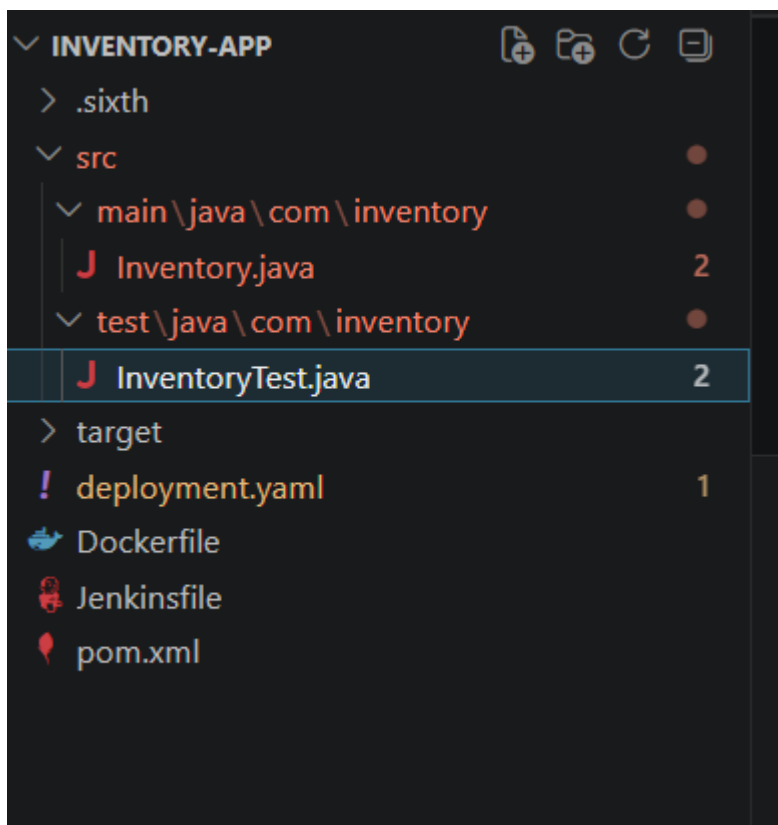


Inventory management system

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1,MAIN JAVA FILE

src/main/java/com/inventory/Inventory.java

```
package com.inventory;
```

```
public class Inventory {
```

```
    public static String checkStock(int quantity) {  
        if (quantity <= 0) return "Out of Stock";  
        else if (quantity <= 10) return "Low Stock";  
        else return "In Stock";  
    }
```

```
public static void main(String[] args) throws Exception {
    System.out.println("Inventory System Running...");

    while (true) {
        Thread.sleep(10000);
    }
}
}
```

2. TEST FILE

 src/test/java/com/inventory/InventoryTest.java

```
package com.inventory;
```

```
import static org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.api.Test;
```

```
public class InventoryTest {
```

```
    @Test
    void testOutOfStock() {
        assertEquals("Out of Stock", Inventory.checkStock(0));
    }
```

```
    @Test
    void testLowStock() {
        assertEquals("Low Stock", Inventory.checkStock(5));
    }
```

```
    @Test
    void testInStock() {
        assertEquals("In Stock", Inventory.checkStock(20));
    }
}
```

3. pom.xml

 pom.xml

```

<project xmlns="http://maven.apache.org/POM/4.0.0">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.inventory</groupId>
  <artifactId>inventory-app</artifactId>
  <version>1.0</version>

  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
  </properties>

  <dependencies>
    <dependency>
      <groupId>org.junit.jupiter</groupId>
      <artifactId>junit-jupiter</artifactId>
      <version>5.10.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-jar-plugin</artifactId>
        <version>3.3.0</version>
        <configuration>
          <archive>
            <manifest>
              <mainClass>com.inventory.Inventory</mainClass>
            </manifest>
          </archive>
        </configuration>
      </plugin>
    </plugins>
  </build>
</project>

```

Dockerfile

```
FROM openjdk:17
WORKDIR /app
COPY target/inventory-app-1.0.jar app.jar
ENTRYPOINT ["java", "-jar", "app.jar"]
```

5. deployment.yaml

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: inventory-app-deployment

spec:
  replicas: 2

  selector:
    matchLabels:
      app: inventory-app

  template:
    metadata:
      labels:
        app: inventory-app

    spec:
      containers:
        - name: inventory-app
          image: santhaseelanjs/inventory-app:latest
          ports:
            - containerPort: 8080
---
apiVersion: v1
kind: Service
metadata:
  name: inventory-app-service

spec:
```

type: NodePort

selector:

app: inventory-app

ports:

- port: 80

targetPort: 8080

nodePort: 30013

6. Jenkinsfile (WINDOWS)

Jenkinsfile

```
pipeline {
  agent any

  environment {
    DOCKER_IMAGE = "santhaseelanjs/inventory-app:latest"
  }

  tools {
    maven 'Maven-3'
    jdk 'JDK17'
  }

  stages {

    stage('Clone') {
      steps {
        git branch: 'main', url: 'https://github.com/YOUR_USERNAME/inventory-app.git'
      }
    }

    stage('Build & Test') {
      steps {
        bat 'mvn clean test'
      }
    }

    stage('Package') {
```

```
    steps {
      bat 'mvn package'
    }
  }

  stage('Docker Build') {
    steps {
      bat 'docker build -t santhaseelanjs/inventory-app:latest .'
    }
  }

  stage('Docker Push') {
    steps {
      bat 'docker push santhaseelanjs/inventory-app:latest'
    }
  }

  stage('Deploy') {
    steps {
      bat 'kubectl apply -f deployment.yaml'
    }
  }
}
```

7. COMMANDS TO RUN

mvn clean package

docker build -t santhaseelanjs/inventory-app .

docker push santhaseelanjs/inventory-app

kubectl apply -f deployment.yaml

J Inventory.java 2 X

J InventoryTest.java 2

📌 pom.xml

🐳 Dockerfile

! d

src > main > java > com > inventory > J Inventory.java > 🐞 Inventory

```
1  package com.inventory;
2
3  public class Inventory {
4
5      public static String checkStock(int quantity) {
6          if (quantity <= 0)
7              return "Out of Stock";
8          else if (quantity <= 10)
9              return "Low Stock";
10         else
11             return "In Stock";
12     }
13
14     Run | Debug
15     public static void main(String[] args) throws Exception {
16         System.out.println(x: "Inventory System Running...");
17
18         while (true) {
19             Thread.sleep(millis: 10000);
20         }
21     }
```

J Inventory.java 2 X

J InventoryTest.java 2 X

📍 pom.xml

🐳 Dockerfile

src > test > java > com > inventory > J InventoryTest.java > 🐞 InventoryTest

```
1  package com.inventory;
2
3  import static org.junit.jupiter.api.Assertions.assertEquals;
4  import org.junit.jupiter.api.Test;
5
6  public class InventoryTest {
7
8      @Test
9      void testOutOfStock() {
10         assertEquals("Out of Stock", Inventory.checkStock(0));
11     }
12
13     @Test
14     void testLowStock() {
15         assertEquals("Low Stock", Inventory.checkStock(5));
16     }
17
18     @Test
19     void testInStock() {
20         assertEquals("In Stock", Inventory.checkStock(20));
21     }
22 }
```


J Inventory.java 2

J InventoryTest.java 2

🔥 pom.xml

🐳 Dockerfile

```
! deployment.yaml > {} spec > {} template > {} spec > [] containers > {} 0 > abc name
io.k8s.api.core.v1.Service (v1@service.json) | io.k8s.api.apps.v1.Deployment (v1@deployment)
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: inventory-app-deployment
5
6  spec:
7    replicas: 2
8
9    selector:
10     matchLabels:
11       app: inventory-app
12
13     template:
14       metadata:
15         labels:
16           app: inventory-app
17
18       spec:
19         containers:
20           - name: inventory-app
21             image: santhaseelanjs/inventory-app:latest
22             ports:
23               - containerPort: 8080
24     ---
25  apiVersion: v1
```

🐳 Dockerfile > ...

```
1  FROM openjdk:17
2  WORKDIR /app
3  COPY target/inventory-app-1.0.jar app.jar
4  ENTRYPOINT ["java", "-jar", "app.jar"]
```

Jenkinsfile

```
1  pipeline {
13    stages {
14      stage('Build & Test') {
20
21        steps {
22          bat 'mvn clean test'
23        }
24      }
25    }
26
27    stage('Package') {
28      steps {
29        bat 'mvn package'
30      }
31    }
32
33    stage('Docker Build') {
34      steps {
35        bat 'docker build -t santhaseelanjs/inventory-app:latest .'
36      }
37    }
38
39    stage('Docker Push') {
40      steps {
41        bat 'docker push santhaseelanjs/inventory-app:latest'
42      }
43    }
44  }
```

```
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ inventory-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.inventory.InventoryTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.064 s -- in com.inventory.InventoryTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ inventory-app ---
[INFO] Building jar: A:\agile\real fat\inventory-app\target\inventory-app-1.0.jar
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.452 s
[INFO] Finished at: 2026-04-13T11:14:54+05:30
[INFO]
[INFO] -----
[+] Building 6.0s (1/2) docker:
```

```
pom.xml
1  <project xmlns="http://maven.apache.org/POM/4.0.0">
2      <artifactId>inventory-app</artifactId>
3      <groupId>com.inventory</groupId>
4      <version>1.0</version>
5
6      <properties>
7          <maven.compiler.source>17</maven.compiler.source>
8          <maven.compiler.target>17</maven.compiler.target>
9      </properties>
10
11      <dependencies>
12          <dependency>
13              <groupId>org.junit.jupiter</groupId>
14              <artifactId>junit-jupiter</artifactId>
15              <version>5.10.1</version>
16              <scope>test</scope>
17          </dependency>
18      </dependencies>
19
20      <build>
21          <plugins>
22              <plugin>
23                  <groupId>org.apache.maven.plugins</groupId>
24                  <artifactId>maven-jar-plugin</artifactId>
25                  <version>3.3.0</version>
26                  <configuration>
27                      <archive>
28                          <manifest>
29                              <mainClass>com.inventory.Inventory</mainClass>
30                          </manifest>
31                      </archive>
32                  </configuration>
33              </plugin>
34          </plugins>
35      </build>
36  </project>
```

PROBLEMS 5 OUTPUT TERMINAL PORTS DEBUG CONSOLE POSTMAN CONSOLE

Containers [Give feedback](#)

Container CPU usage 9.71% / 1600% (16 CPUs available) Container memory usage 827.12MB / 7.26GB [Show charts](#)

Search Only show running containers

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last start...	Actions
☐	k8s_POD_inventory-app-d	e731d924aec5	pause:3.10		0%	5 minutes ago	
☐	k8s_POD_inventory-app-d	1500399a41ab	pause:3.10		0%	5 minutes ago	



Status

Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

JENKINS PIPELINE (INVENTORY SYSTEM)

Permalinks

Builds >

Today

#1 11:16 AM   



Configure

General

Triggers

Pipeline

Advanced

☐ Trigger builds remotely (e.g., from scripts) ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
37 }
38
39 stage('Docker Push') {
40   steps {
41     bat 'docker push santhaseelanjs/inventory-app:latest'
42   }
43 }
44
45 stage('Deploy to Kubernetes') {
46   steps {
47     bat 'kubectl apply -f deployment.yaml'
48   }
49 }
50 }
```

try sample Pipeline...

Save

Apply

Jenkins

JENKINS PIPELINE (INVEN... / #1 / Stages

Q

Graph

Start

Tool Install

Clone

Build & Test

Package

Docker Build

Docker Push

Deploy to Kubernetes

End

Q Search

Clone

3s

Started 31s ago

Jenkins

✓ Tool Install 0.14s

✗ Clone 3s

➤ Build & Test 38ms

➤ Package 28ms

➤ Docker Build 32ms

➤ Docker Push 30ms

➤ Deploy to Kubernetes 27ms

✓ Use a tool from a predefined Tool Installation [Maven-3](#) > 23ms

✓ Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step. > 29ms

✓ Use a tool from a predefined Tool Installation [JDK17](#) > 24ms

✓ Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step. > 33ms

✗ Git > 3s

0 The recommended git tool is: NONE

1 No credentials specified

2 Cloning the remote Git repository

3 Cloning repository https://github.com/YOUR_USERNAME/inventory-app.git

↑

↓